

Workshop 4: Advanced SQL and JOINS

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In this workshop, you will work in a team on a table with many other students around SQL. It is a collective discussion to find a solution to the given problem in the database AdventureWorks2019 (Figure 2), step by step by exchange and comments. Because we will need a **connection to MS SQL Server**, you will have to test the solutions from Canvas for Windows and MacOS computers (is the MacOS solution working under Linux?). Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue...

1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

2 Computing queries (10 min.)

Please answer to the following questions together with your colleagues around the table. You can write down the answers in a shared document so that everyone around the table can get the correct solutions, but you can also just discuss and exchange ideas. The most important is to have a discussion and to find a common answer together.

- Using JTH MSSQL Server, compute:
 - The sum of 12345 and 43210.
 - The product of 38 and 17.
 - The division of 10125 by 25.
 - The difference between 135.28 and 33.46.
 - The sine of $\pi/2$.
 - The cosine of $\pi/3$.

3 Basic queries (15 min.)

When the information you are looking for is in one table **ONLY**, no problem, use a simple select to get this information (no need to join any tables).

Please answer to the following questions together with your colleagues around the table. You can write down the answers in a shared document so that everyone around the table can get the correct solutions, but you can also just discuss and exchange ideas. The most important is to have a discussion and to find a common answer together.

- Using JTH MSSQL Server, find:
 - The FirstName and LastName of the persons whose FirstName is "Jeremy".
 - The number of persons whose FirstName is "Jeremy".
 - The FirstName and LastName of the persons whose FirstName ends with "der".
 - The number of persons whose LastName ends with "der".
 - The list of all departments in the company with their GroupName.
 - The JobTitle and the number of employees with this JobTitle.
 - The Jobtitle and the number of employees with a JobTitle containing the text "President".

4 Join queries with two tables (15 min.)

When the information you are looking for is in **MORE THAN ONE** table, use a JOIN select to get this information. When you write your join, identify a **PATH** from the first table to the last one, include them into the JOIN and specify the **common attributes** (with **ON**) between the tables (the foreign keys and primary keys), and FOLLOW YOUR PATH.

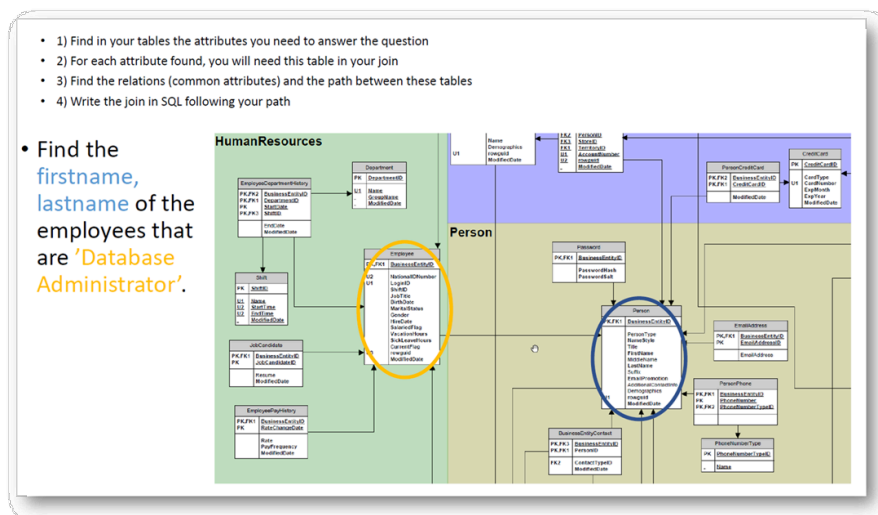


Figure 1: A join between two tables.

- Discuss with your colleagues around the table about the following questions (and please answer them together). Using JTH MSSQL Server, find:
 - The ProductID, Name, Color, StartDate, Enddate, and ListPrice for each product in the table Product.
 - The name, minimal price, maximum price, average price for each product in the table Product and order them by the average price, in descending order.

5 Join queries with three tables (15 min.)

When the information you are looking for is in MORE THAN TWO tables, use a JOIN select to get this information. When you write your join, identify a PATH from the first table to the last one, include them into the JOIN and specify the **common attributes** (with **ON**) between the tables (the foreign keys and primary keys), and **FOLLOW YOUR PATH**.

- Discuss with your colleagues around the table about the following questions (and please answer them together). Using JTH MSSQL Server, find:
 - The FirstName, LastName, CreditCard.CardNumber for each person in the table Person.
 - The FirstName, LastName, CreditCard.CardNumber for the person that has the CardNumber 55559523560296.
 - The FirstName, LastName, CreditCard.CardNumber for whom the CardNumber ends with 2444.
 - The JobTitle and GroupName of all the employees working in the department whose GroupName is 'Research and Development'.

6 Join queries with more than three tables (15 min.)

- Discuss with your colleagues around the table about the following questions (and please answer them together). Using JTH MSSQL Server, find:
 - The FirstName, Lastname, JobTitle, Department Name, StartDate, EndDate of the employee whose name is 'Laura Norman' (write the SQL query from Person to Department).
 - The FirstName, Lastname, JobTitle, Department Name, StartDate, EndDate of the employee whose name is 'Laura Norman' (write the SQL query from Department to Person).

7 If you are done...

In case you are done with the workshop very fast, you can work as a group on the next lab assignment. :) B+

